

Jinmin Kim (김진민)

CURRICULUM VITAE

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Portfolio: <https://rafaam11.github.io> | GitHub: github.com/rafaam11

Summary

Robotics and computer vision R&D engineer (2023.02–) developing optical 3D tracking, coordinate transformation, 3D registration, surgical navigation, XR and robotic system integration. Researched computer-assisted surgery and registration at DGIST, and 3+ years of collaborative R&D with Samsung Medical Center oral and maxillofacial surgery on digital surgical planning, XR consultation and image-guided navigation.

Research interests: Surgical Robotics · Physical AI · Surgical AI · 3D Vision · Robot Perception · Multimodal Perception · Spatial Intelligence

Education

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

2021.03 – 2023.02

M.S. in Robotics and Mechatronics Engineering

- Surgical Robotics and AR Laboratory (advisor: Jaesung Hong) — surgical navigation team
- Thesis: Virtual Jawbone Reduction Planning based on Dental Occlusion for Mandibular Fracture Surgery
- Oral and maxillofacial AR surgery programme (2021.06–2023.05) — early and supporting AR navigation development

Kumoh National Institute of Technology

2015.03 – 2021.02

B.S. in Mechanical System Engineering

- Undergraduate researcher, System & Vision Lab. (2019–2020)
- Completed the DGIST and UNIST summer undergraduate research programmes
- Five-bar-linkage quadruped robot (2020.04–2021.01) — mechanism design, kinematic modelling, and control; led to a patent application and an engineering-fair award
- Vice president of an invention club (2019) — patent applications and invention-contest awards; designated an outstanding education-donation club by the Ministry of Education

Professional Experience

DIGITRACK Inc. — Research Engineer, Software R&D

2023.02 – Present

DGIST spin-off developing high-accuracy optical 3D tracking systems, surgical navigation software, and robotic/automation solutions, Daegu

1) Optical 3D Tracking Software

- The in-house optical 3D tracker **SKADI: API and Viewer software development and maintenance** — optical marker tracking, calibration, coordinate transformation, 3D visualisation and surgical tool tracking; the software layer of a product line delivered to Korean surgical-navigation companies and research institutes

2) Surgical Navigation & Medical R&D — long-term collaboration with Samsung Medical Center

- **Long-term collaborative R&D with the Department of Oral and Maxillofacial Surgery, Samsung Medical Center (3+ years)** — bi-weekly clinical research meetings since 2023; main engineer on two government programmes and one development contract; sole developer of the 3D Slicer-based **integrated surgical navigation software used at the hospital** (CT–patient–tracker–instrument registration and calibration, six maxillofacial surgical workflows, HoloLens 2 and Meta Quest integration)
- **Oral and maxillofacial digital surgical workflow** — connected surgical planning → AI-assisted planning → XR consultation → surgical navigation as stage-by-stage software

- **Technical lead, digital occlusion software (2026.03–)** — Custom App architecture, UI/UX, landmark and occlusion engines; integrated an AI-based 3D anatomical landmark detection model into planning and registration; researcher validation in progress
- **Led the rTMS coil-navigation system (NeuroPilot)** — digital-twin visualisation of coil pose (government health project)
- **Low-dose C-arm 3D navigation** — linked C-arm images to the 3D navigation workflow (owned scope)
- **Surface-guided respiratory tracking and 4D CT registration** (large KERI programme, 2026.06–)
- Delivered several development contracts for research institutes including KERI and KIMM

3) Robotics & Vision — Autonomous Forklift (2025.11–)

- Extending optical tracking and coordinate registration to robot–camera integration and robot vision
- Designed and implemented the safety-policy manager (safety_manager) for the in-house autonomous-forklift solution (DOTORI); integrated LiDAR and ToF sensors over Zenoh/ROS 2
- **Own vision R&D** — truck-loading vision (ToF+RGB, SAM-family segmentation PoC, behaviour-tree integration)

Publications & Presentations

Ahn, J., **Kim, J.** (2026). Development and Clinical Evaluation of an Extended Reality-Based Consultation Platform "OMFS VR Consultation App" for Enhanced Patient Understanding in Orthognathic Surgery. *2026 Spring Conference, Korean Society of Medical Metaverse*. (Poster)

Kim, J., Jeung, D., Cho, R., Yang, B., Hong, J. (2024). A Proof of Concept: Optimized Jawbone-Reduction Model for Mandibular Fracture Surgery. *Journal of Imaging Informatics in Medicine*. (Journal article · SCIE, Q1, IF 4.2) — joint first author

Ahn, J., **Kim, J.** (2024). Set-up of Dental Final Occlusion using Haptic Device in Orthognathic Surgery. *2024 Spring Conference, Korean Society of Medical Metaverse*. (Poster · Best Poster Award)

Jeung, D., **Kim, J.**, Cho, R., Yang, B., Hong, J. (2023). Occlusion-based Jawbone Model Optimization for Mandibular Fracture Surgery. *14th Conference, Korean Society of Medical Robotics*. (Oral presentation · Best Paper Award)

Kim, J. et al. (2022). Dental Occlusion Model Using Arch Line for Mandibular Fracture Surgery. *ACCAS 2022*, Bangkok, Thailand. (Conference paper)

Kim, J., Jeung, D., Cho, R., Yang, B., Hong, J. (2022). Virtual Jawbone Model Reconstruction for Mandibular Fracture Surgery Guidance. *13th Conference, Korean Society of Medical Robotics*. (Oral presentation)

Kim, J., Kim, Y., Lee, J., Park, K. (2019). Design of Ping-Pong Ball Launcher. *ISM 2019 — International Symposium on Mechatronics*. (Conference paper)

Patents — 7 applications · 3 granted

Employment (DIGITRACK)

- [Granted] Coordinate-frame registration for real-time 3D surgical-tool tracking (10-2024-0186869, 2024.12)
- [Granted] Position tracking device and method (10-2024-0186864, filed 2024.12)
- [Filed] Two-axis rotating stereo-camera actuation device and method (10-2024-0127937, 2024.09)

Additional patents (undergraduate)

- [Filed] Quadruped walking robot using a multi-joint link structure (10-2021-0008332, 2021.01) · [Granted] Clothes hanger with an elastic section (10-2019-0100328, 2019.08) · [Filed] Assemblable carrier with adjustable storage space (10-2019-0010185, 2019.01) · [Filed] Disposable paper-cup collection bin (10-2015-0122661, filed 2015.08)

Honors & Awards

- Best Poster Award (2024, Korean Society of Medical Metaverse)
- Best Paper Award (2023, Korean Society of Medical Robotics)
- Undergraduate: Grand Prize, International TRIZ Competition (2019) · Grand Prize, Startup Idea Competition (2019, decontamination spray) · Encouragement Prize, KIT Engineering Fair (2020, five-bar-linkage quadruped robot)
- Undergraduate: Invention Contest Sponsor's Award (2019) · Excellence Award (2015) · Hyosung GREEN Earth Excellence Award (2019) · Bronze Prize, ROS training (2020)
- Undergraduate: Outstanding education-donation club, Ministry of Education (2019, club vice president)

Skills & Languages

Surgical navigation & imaging: 3D Slicer, DICOM, optical tracking, coordinate transformation, calibration, image/point-cloud registration, OpenIGTLink

Computer vision & 3D: point clouds, rigid/ICP registration, segmentation (SAM family), 3D visualisation, PyTorch

Robotics: ROS 2, Zenoh, Isaac Sim, behaviour trees, vision and sensor integration

XR & spatial computing: Unity, OpenXR, MRTK, HoloLens 2, Meta Quest, Galaxy XR

Programming: Python, C++, C#

Languages: English — intermediate (paper writing and reading) · Japanese — JLPT N2 (2021)